Treasury has issued $739 billion of TIPS, which represents 7.5% of all outstanding marketable Treasury securities.[5] Officially, these bonds are now referred to as *Treasury Inflation Indexed Securities*, but market participants and all the literature, except for official government publications, continue to use the name TIPS.

Many sovereigns have issued real bonds. The first linker was issued by Massachusetts in 1780 to raise money for the Revolutionary War.[6] Inflation was extremely high while patriots were fighting the British, and the bonds were invented partly to allay the anger of soldiers in the revolutionary army who were dismayed by the declining purchasing power of their (already meager) pay. An important linkers market is in Great Britain, where the Bank of England first issued *Inflation-Linked Gilts* in 1981. This market was immediately successful because of high inflation in the United Kingdom during that time (significantly higher than U.S. inflation, and the Great Inflation lasted longer in the United Kingdom than in the United States). The U.S. market, in contrast, was highly illiquid for the first few years (see below). Linkers now account for approximately a quarter of all outstanding debt issued by the U.K. government.

TIPS pay a fixed coupon payment that is indexed to CPI with a lag of three months.[7] The interest rate on TIPS is fixed, but the bond's principal is regularly adjusted for inflation, with the result that investors receive increasing interest payments when inflation rises. The principal can also fall with inflation, but it can never go below the original face value of the bond. This *deflation put* usually has little value for a long-maturity TIPS bond that has a high principal balance. Such a bond has already experienced a period of high inflation. For this bond, deflation would have to be extremely severe for its principal to fall back to par. In contrast, during times of low inflation and deflation, the deflation put is valuable for short-maturity TIPS.[8] These bonds have low principal balances, and, when deflation occurs, their principals cannot go below par. Such a period occurred during 2008 and 2009. Inflation during parts of 2009 was actually negative (see Figure 11.1 and 11.3), and newly issued TIPS during that time were extra valuable because of the high value in their deflation puts.

4.1. REAL BONDS FOR RETIREMENT?

An investor buying and holding a TIPS bond when it is first issued receives a constant real yield due to the inflation indexation.[9] If the TIPS bond is held to pay

[5] Monthly Statement of the Public Debt of the United States, Dec. 31, 2011.

[6] See Humphrey (1974) and Shiller (1993).

[7] Since CPI is released with a two-week delay, there is actually a two and a half month lag between the actual release of CPI and the indexation adjustment.

[8] Jacoby and Shiller (2008) show that the value of the deflation put can be large.

[9] Well, not quite. There is still the reinvestment risk associated with the coupons of the bond. One would technically need to buy and hold a TIPS zero-coupon bond to receive a constant real yield. The text ignores this consideration.